Operating Systems

Unit 4 – Orange Program Assignment

Name : Shreya C Rao

SRN : PES2UG24CS481

SEC : 4 H

SCREENSHOTS:

1A)

```
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$
```

1B)

```
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$
```

2A)

```
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$
```

2B)

```
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$ xxd .pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442 | head -20
00000000: 7472 6565 2034 3900 3130 3036 3434 2068  tree 49.100644 h
00000010: 656c 6c6f 2e74 7874 0066 2246 63d2 3e6f  ello.txt.f"Fc.>o
00000020: 4d9d e9e2 c7e6 d876 4305 a92a 3830 a1a5  M......vC..*80..
00000030: 2d3d 5f4a a800 7b5c 39                   -=_J..{\9
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$
```

3A)

```
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$ make pes
./pes init

echo "hello" > file1.txt
echo "world" > file2.txt

./pes add file1.txt file2.txt
./pes status
make: 'pes' is up to date.
Initialized empty PES repository in .pes/
Staged changes:
  staged: file1.txt
  staged: file2.txt
```

3B)

```
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776268966 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776268966 6 file2.txt
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$
```

4A)

```
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$ ./pes init

echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"

echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Bye" > bye.txt
./pes add bye.txt
./pes commit -m "Add bye"

./pes log
Initialized empty PES repository in .pes/
Committed: cfdb4bb23eb9... Initial commit
Committed: cfdb4bb23eb9... Initial commit
Committed: 7574fa4a65da... Add world
Committed: 7574fa4a65da... Add world
Committed: a7dba38061a0... Add bye
Committed: a7dba38061a0... Add bye
commit a7dba38061a07ce723b59eb721f1d554f9352766f5fb99399081bdef9fb05114
Author: Shreya Rao <PES2UG24CS481>
Date:   1776275093

    Add bye

commit 7574fa4a65daea10360c93ba6e8b29502ee4baa045abf2eddfbde29945ba7d51
Author: Shreya Rao <PES2UG24CS481>
Date:   1776275093

    Add world

commit cfdb4bb23eb92fdf2e73e3fb87fe82872808f4cbbc73e6f7edc41b562e5bbfd8
Author: Shreya Rao <PES2UG24CS481>
Date:   1776275093

    Initial commit
```

```
commit 86530d76fcc4d0fda36ddc6c91b030577bcff6c6ca277fce87a09ebc699babf4
Author: Shreya Rao <PES2UG24CS481>
Date:   1776273509

    Third commit

commit d851f69f548a74097fffa13ec24b1f6239088a7fb17ec05a7526d1de083cfbb4
Author: Shreya Rao <PES2UG24CS481>
Date:   1776273509

    Second commit

commit bbb8927575105c1ffb581288bef6f6a28f3208754692f8a3d02ae2b7e431d2d6
Author: Shreya Rao <PES2UG24CS481>
Date:   1776273509

    Initial commit
```

```
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$ ./pes log
commit a7dba38061a07ce723b59eb721f1d554f9352766f5fb99399081bdef9fb05114
Author: Shreya Rao <PES2UG24CS481>
Date:   1776275093

    Add bye

commit 7574fa4a65daea10360c93ba6e8b29502ee4baa045abf2eddfbde29945ba7d51
Author: Shreya Rao <PES2UG24CS481>
Date:   1776275093

    Add world

commit cfdb4bb23eb92fdf2e73e3fb87fe82872808f4cbbc73e6f7edc41b562e5bbfd8
Author: Shreya Rao <PES2UG24CS481>
Date:   1776275093

    Initial commit

commit 86530d76fcc4d0fda36ddc6c91b030577bcff6c6ca277fce87a09ebc699babf4
Author: Shreya Rao <PES2UG24CS481>
Date:   1776273509

    Third commit

commit d851f69f548a74097fffa13ec24b1f6239088a7fb17ec05a7526d1de083cfbb4
Author: Shreya Rao <PES2UG24CS481>
Date:   1776273509

    Second commit

commit bbb8927575105c1ffb581288bef6f6a28f3208754692f8a3d02ae2b7e431d2d6
Author: Shreya Rao <PES2UG24CS481>
Date:   1776273509

    Initial commit
```

4B)

```
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b5253611147faa7b5b922919e9a23143d
.pes/objects/0f/8d59d2d3c9c97b7be5f22dc0e38173f9059f09af6ac845de19fb4ef8e90731
.pes/objects/23/a0830d6ae825928efdedabe18a35faf7f7074f19d83d9dd4ed75ee555584dd
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/3b/643ad70527b6958e0a2c29a950496617c355932e25571de833b00b76b539e3
.pes/objects/48/1407f7237f7bd749dca31e2d8422bafe211741d5556f007e0faf821ffc7179
.pes/objects/52/35aa78924f1785a7bf34a3058cffb0225b390d3c1ebc9c611cd58e8d4965b4
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/6e/1561fa172d8a928031a9cc5ab7fec83243c9801855315d3fc8ab72f102288b
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/8d/e661a3024bd0ad06fb91b9c0d2f692136479a234faebf1b41bdadede1c08a5
.pes/objects/8e/a3385ad5ba7f3ec4151ace1ce063d3da36a33eda38e30b67a515f4526d9f20
.pes/objects/cd/2140b058cabb1e059848a020f871a6e9e8fe6d7ba257fffbac45127600a715
.pes/objects/dd/9c5c595496baa1e59cfeeb412f0e228af4ecd9f648ed20d4ceca4e4c2060a8
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/refs/heads/main
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$
```

4C)

```
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
8ea3385ad5ba7f3ec4151ace1ce063d3da36a33eda38e30b67a515f4526d9f20
ref: refs/heads/main
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$
```

Branching and Checkout

Q5.1: A branch in Git is just a file in .git/refs/heads/ containing a commit hash. Creating a branch is creating a file. Given this, how would you implement pes checkout <branch> — what files need to change in .pes/, and what must happen to the working directory? What makes this operation complex?

- To check out a branch, three things must change in .pes/: update HEAD to contain ref: refs/heads/<branch>, then read the commit hash from that branch file, then load the tree the commit points to.
- For the working directory, you must recursively walk the target tree, reading each blob from the object store and overwriting the corresponding file on disk.
- New files in the target not present currently must be created; files tracked in the current branch but absent in the target must be deleted. The complexity comes from the fact that the working directory is a live filesystem — you must be careful about partial failures (e.g., disk full), and must handle nested subdirectories by creating them in order.
- Also, new binary or permission state must be restored from tree entry modes.

Q5.2: When switching branches, the working directory must be updated to match the target branch's tree. If the user has uncommitted changes to a tracked file, and that file differs between branches, checkout must refuse. Describe how you would detect this "dirty working directory" conflict using only the index and the object store.

- For each file in the index, compare its recorded mtime and size against the actual file on disk (fast path).
- If they differ, re-hash the file and compare with the stored blob hash (slow path). If the hash differs from what the index recorded, the file has local uncommitted changes.
- Then check whether that same file path exists in the target branch's tree with a *different* blob hash.

- If both conditions hold — local modification AND the target branch has a different version — checkout must be refused. The entire process requires only the index and the object store; no diff tool is needed.

Q5.3: "Detached HEAD" means HEAD contains a commit hash directly instead of a branch reference. What happens if you make commits in this state? How could a user recover those commits?

- In detached HEAD state, HEAD contains a raw commit hash rather than ref: refs/heads/main. Commits made in this state are recorded in the object store and each points to the previous, forming a real chain — but no branch file is updated.
- If you switch away without creating a branch, these commits become "unreachable" (no branch points to them).
- To recover, you can run git reflog (which logs every HEAD change including detached commits), find the hash of the last commit made, then create a branch at that hash: git branch recovered-work <hash>.
- In PES-VCS without reflog, you'd need to scan all objects for commits whose parent chain doesn't trace back through any known branch — essentially a reverse GC walk.

Garbage Collection and Space Reclamation

Q6.1: Over time, the object store accumulates unreachable objects — blobs, trees, or commits that no branch points to (directly or transitively). Describe an algorithm to find and delete these objects. What data structure would you use to track "reachable" hashes efficiently? For a repository with 100,000 commits and 50 branches, estimate how many objects you'd need to visit.

- Start from every branch ref and HEAD, collect the commit hash. From each commit, collect its tree hash. From each tree, recursively collect all blob and subtree hashes.
- Use a hash set (e.g., a C hash table or sorted array of ObjectIDs) to store all "reachable" hashes. Then find .pes/objects -type f and for each object file, reconstruct its hash from its path — if it's not in the reachable set, delete it.
- For a repo with 100,000 commits averaging 20 unique objects each, you'd need to visit roughly 2 million objects. With 50 branches, the reachable set might cover 1–1.5 million objects, leaving ~500K to delete.

Q6.2: Why is it dangerous to run garbage collection concurrently with a commit operation? Describe a race condition where GC could delete an object that a concurrent commit is about to reference. How does Git's real GC avoid this

- Suppose GC runs find .pes/objects and builds its reachable set snapshot at time T.
- Concurrently, a commit operation at T+1 writes a new blob and is about to write a tree that references it, but hasn't updated HEAD yet.
- GC sees the blob as unreachable (nothing points to it yet) and deletes it. The commit operation then writes a tree pointing to the now-deleted blob — the object store is corrupt.
- Git avoids this by using a "grace period" — objects newer than 2 weeks are never deleted, even if unreachable.
- This means a concurrent commit's objects are always young enough to survive GC. Additionally, Git's GC holds a lock file (.git/gc.pid) so only one GC runs at a time.

FINAL:

```
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged: file.txt
  staged: hello.txt

--- First Commit ---
Committed: 3efbb352ff55... Initial commit

Log after first commit:
commit 3efbb352ff5582f7641fa6ea34e4d05790828a97b58ab128489c97fc091021f1
Author: PES2UG24CS481
Date:   1776270810

    Initial commit


--- Second Commit ---
Committed: 421846644203... Update file.txt

--- Third Commit ---
Committed: 6904f71245b2... Add farewell

--- Full History ---
commit 6904f71245b259cf0e9f4be59a29749de00234670c980c30f81600939c02d1af
Author: PES2UG24CS481
Date:   1776270810

    Initial commit
```

```
--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
6904f71245b259cf0e9f4be59a29749de00234670c980c30f81600939c02d1af

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/3e/fbb352ff5582f7641fa6ea34e4d05790828a97b58ab128489c97fc091021f1
.pes/objects/42/1846644203b5d958e94c46ab3a143e10bf3882806bac2473d22cc61afc868b
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/69/04f71245b259cf0e9f4be59a29749de00234670c980c30f81600939c02d1af
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
shreya@SHREYACRAO:~/PES2UG24CS481-pes-vcs$
```